QP CODE: 25900802

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)REGULAR/ IMPROVEMENT/ REAPPEARANCE EXAMINATIONS, NOVEMBER 2025
First Semester
Discipline Specific COre Course (DSC) - MG1DSCECT101- COMPUTER FUNDAMENTALS AND BASICS OF PC HARDWARE
(2024 Admission Onwards)

Duration: 1.5 Hours **Max.Marks: 50**

PART A
Multiple Choice Questions.
Answer All Questions.
*Each Question carries **1** mark.*

1. c). ENIAC [U] / [CO1]
2. b). Terabyte [U] / [CO2]
3. d). 8,400 MT/s [U] / [CO3]
4. d). For more compact representation of large binary numbers. [U] / [CO1]
5. c). RAM [K] / [CO2]
6. d). SSDs use less power, improving battery life in portable devices. [An] / [CO3]
7. b). Limited to a small geographic area, typically within a building [U] / [CO1]
8. b). SSDs have no moving parts, making them more durable and energy efficient. [U] / [CO2]
9. c). Executing instructions. [U] / [CO1]
10. b). Image sensor (CCD or CIS) [U] / [CO2]

(10 x 1 = 10)

PART B
Short Answer Type Questions.
*Answer any **4** Questions.*
*Each Question carries **5** marks.*

11. DC Regulated Power Supply
    Components - Step-down transformer, Rectifier, DC filter, Voltage regulator
    Block Diagram - 2 Marks
    List of Components and Explanation - 3 Marks [U] / [CO2]

12. Differences between inkjet and laser printers
    Inkjet printers use liquid ink sprayed through tiny nozzles to produce high-quality color images, while laser printers use toner powder and a laser beam to create fast, sharp, and smudge-proof text documents.

Any 5 differences - 1 Mark each (Total - 5 Marks) [U] / [CO2]

13. Different Types of memory technologies used in GPUs
    - DDR Memory
    - GDDR Memory
    - High Bandwidth Memory (HBM)
    - LPDDR(Low Power DDR) Memory

    Explanation of these memory technologies - 5 marks [An] / [CO3]

14. Green Computing can help reduce environmental impact
    Green computing definition - 1 mark
    Two major ways - 2 marks each
        (i). Improving energy efficiency
        (ii). E-waste reduction [An] / [CO3]

15. Types and importance of cooling systems in computers
    Importance of Cooling Systems - Prevents Overheating, Improves Performance, Extends Component Lifespan, Ensures System Stability, Supports Overclocking, Energy Efficiency
    Any 5 importances - 2.5 Marks
    Types of cooling systems - Air cooling, Liquid Cooling, Passive Cooling, Thermoelectric Cooling, Hybrid Cooling, Immersion Cooling
    Any 2 types - 2.5 Marks [U] / [CO2]

16. Cache memory improves performance in multi-core processors by providing
    Faster Data Access, Reduced Memory Latency, Improved Parallel Processing, Shared Cache for Coordination (L3 Cache), Lower Bus Traffic
    Explanation - 5 Marks [U] / [CO2]

(4 x 5 = 20)

PART C

Essay Type Questions
*Answer any **2** Questions.*
*Each Question carries **10** marks.*

17. Types of operating Systems
    - Batch Operating System
    - Multi-Programming System
    - Multi-Processing System
    - Multi-Tasking Operating System
    - Time-Sharing Operating System
    - Distributed Operating System
    - Network Operating System
    - Real-Time Operating System

Explanation of any 5 Operating systems with its characteristics, advantages and disadvantages - 2 Marks each (Total - 10 Marks) [U] / [CO1]

18. Types of USB ports - USB Type A, USB Type B, USB Type C, USB Micro A, Micro B, Mini USB
    Explanation of any 4 types - 1 Mark each (Total - 4 Marks)
    Types of CPU Sockets - Pin Grid Array (PGA), Land Grid Array (LGA), Ball Grid Array (BGA)
    Explanation of 3 types - 2 Marks each (Total - 6 Marks)
    [U] / [CO2]
19. Neuromorphic Computing
    Neuromorphic computing is an advanced field of computer engineering that aims to design hardware and systems that mimic the structure and functioning of the human brain.
    Definition - 1 Mark
    Key Concepts - 4 Marks
    Advantages - 2 Marks
    Disadvantages - 2 Marks
    Applications - 1 Mark [An] / [CO3]

20. Components of a Computer System

Diagram - 2 Marks

Components - Input unit, Output unit, Storage unit - Primary Memory and Secondary Memory, Central Processing unit, ALU, Control Unit)

Explanation of Components - 8 Marks [U] / [CO1]

(2 x 10 = 20)